Trig, Inverse Trig, and Series Review

Notes

Trig Review

This table is a great mnemonic for remembering special angle values.

Table Trick

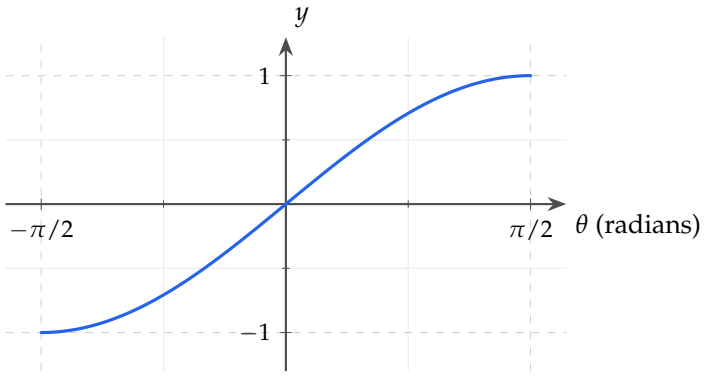
θ	30°	45°	60°	90°
$\sin(\theta)$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(\theta)$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	0
$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Undefined $(\frac{1}{0})$

Notice the pattern: $\sin(\theta)$ counts up ($\sqrt{1}, \sqrt{2}, \sqrt{3}$ over 2), $\cos(\theta)$ counts down, and $\tan(\theta)$ is simply their quotient.

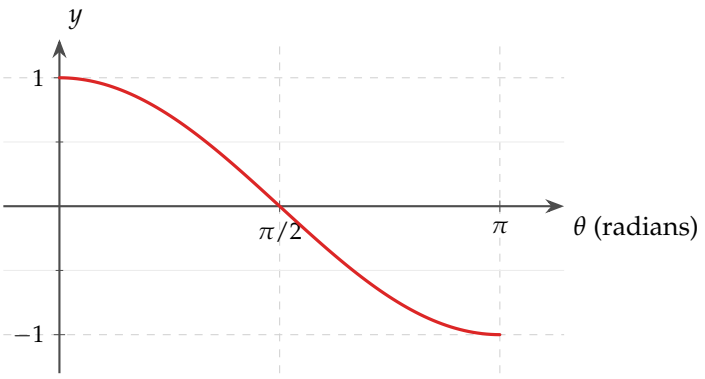
Standard Trig Graphs

Below are the standard trig graphs you are probably used to. You can get these similarly by just reading the table, but it is good to have them down to save time. I am restricting the domain to only $[-\pi/2, \pi/2]$ for a reason. The one exception is $\cos(\theta)$, which is graphed on $[0, \pi]$ instead.

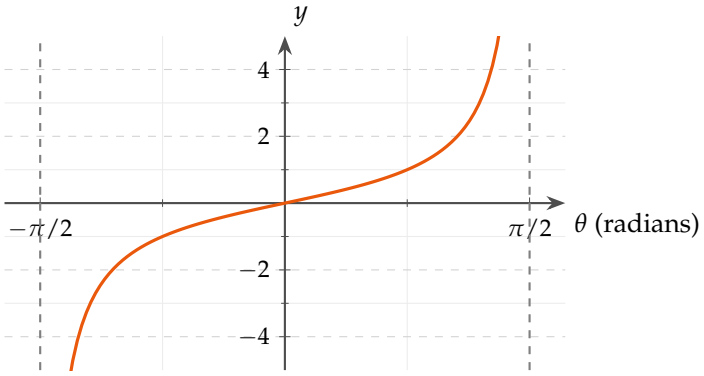
Graph of $y = \sin(\theta)$



Graph of $y = \cos(\theta)$



Graph of $y = \tan(\theta)$

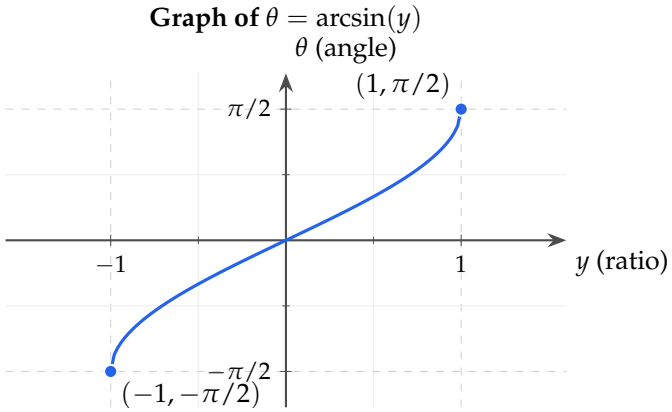


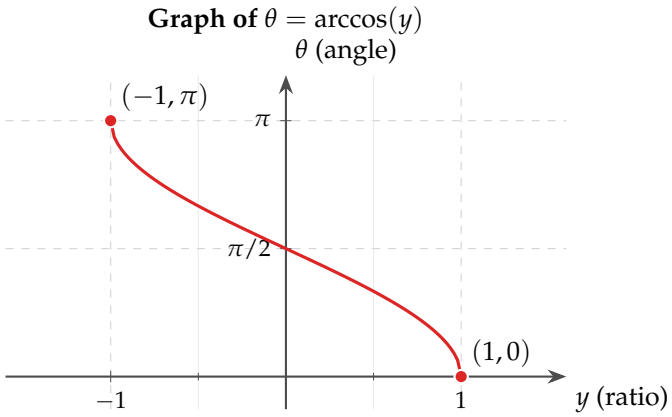
Inverse Trig Graphs

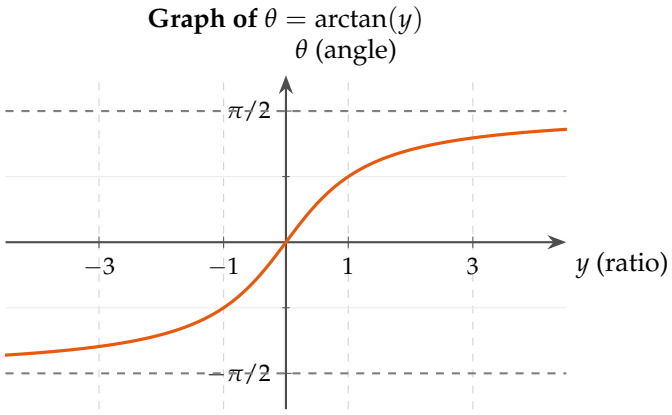
Explanation

An inverse function $y = f^{-1}(x)$ undoes $f(x)$ by swapping the input (x) and output (y). Graphically, this corresponds to reflecting across the line $y = x$. This also makes the domain of $f(x)$ become the range of $f^{-1}(y)$ and range of $f(x)$ becomes the domain of $f^{-1}(y)$. Since a well-defined function requires one output per input, we see that trigonometric functions repeating behavior breaks this. If the output of $\sin(\theta)$ is zero at two points, for instance, then there are multiple choices for the input of $\arcsin(0)$. Therefore, we have to restrict our choices to no repeats. Once we do that, we can graph the plots fairly quickly.

Inverse Trig Plots







Sequences and series

Background and Explanation of Sequences and Series

Explanation

A sequence of terms (a_n) is just an ordered list so like $a_1, a_2, a_3, \ldots$ that follows some rule for computing the a_nth term. Here are examples:

- $a_n = \frac{1}{n}$: $a_1 = 1, a_2 = 1/2, a_3 = 1/3, a_4 = 1/4, \ldots,$
- $a_n = \arcsin(1/n)$: $a_1 = \arcsin 1, a_2 = \arcsin \frac{1}{2}, \ldots$

There are fun questions to ask about these guys: *what value does a_n tend to as n gets larger and larger?* (aka $\lim a_n$?) *What happens if you add each of the terms?* (aka $\sum a_n$?) and so on! The question we are asking is: how does $\sum a_n$ relate to (a_n), or equivalently, how does the series relate to a sequence?
Let's think about it together. Let's say you have a series

$$\sum_{n=1}^{\infty} a_n$$

and the corresponding sequence

$$(a_n)$$

We know that the series is written out as

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

Well, what happens if

$$\lim_{n\to\infty} a_n \neq 0$$

This would mean you are adding up infinitely many non-zero terms. That is, you grow with no bounds or, at least, never settle down to a value, aka, *diverge*. This idea is known as **the divergence test**. Formally:

The Divergence Test: Given a series $\sum_{n=1}^{\infty} a_n$ if $\lim_{n\to\infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ diverges. ***HOWEVER***, if $\lim_{n\to\infty} a_n = 0$ it is inconclusive.

The reason for inconclusivity is that you are not sure *how fast* you reach zero. And, if you are too slow, you would diverge. If you are quick enough you converge.

Series Explanations

Geometric:
Geometric series should intuitively make sense and I am happy to further elaborate in a future meeting or notes. The basic picture is, if $|r| > 1$ then from $a_n \to a_{n+1}$ you are multiplying by a number that will either yield a more negative ($r < -1$) or more positive ($r > 1$) number for each a_n, so it will grow with no bound. But, if $|r| < 1$ each sequential terms decreases since multiplying by a fraction does that, and it is at a relatively quick pace, thus it converges. When it converges, $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$, where a is the first term.

Telescoping Series:
If you have a series like $\sum \frac{1}{n(n+1)}$, you rewrite it using partial fractions:

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

Then, the n-th partial sum S_n telescopes and collapses:

$$S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

Everything in the middle cancels, leaving only the first and last parts:

$$S_n = 1 - \frac{1}{n+1}$$

To find the sum of the infinite series, take the limit as $n \to \infty$:

$$\text{Sum} = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1 - 0 = 1$$

p-series:
If you have a function $f(x) = x^{-p}$ for $p > 1$, the terms go to 0 quickly. In fact, they approach 0 quick enough that the series defined by $\sum a_n$ where $a_n = f(n) = \frac{1}{n^p}$ converges. If $p \leq 1$ (for example $p = 1$, the harmonic series), the terms do not go to 0 quick enough and the series diverges.

Order of Tests

Here is a useful order to do tests. We are observing

$$\sum_{n=1}^{\infty} a_n$$

1. Is it a $p-$series, telescoping series or geometric series? If it is, then you can immediately get your result.

2. Is $\lim_{n\to\infty} a_n \neq 0$? If it is, then you are done. Otherwise, proceed down the list.

3. Let $a_n = f(n)$ and consider the sub-series $\sum_{n=k}^{\infty} a_n$. Is $f(x)$ continuous, positive and decreasing? If it is, if $\int_k^{\infty} f(x)\,dx$ diverges, then so does our series. If it converges, then so does our series.

4. Is there a sequence b_n where you realize $0 \leq b_n \leq a_n$ for all n? If there is then: if $\sum b_n$ diverges then $\sum a_n$ does too. Otherwise, is there c_n where you realize that $0 \leq a_n \leq c_n$ for all n? If there is then: if $\sum c_n$ converges, $\sum a_n$ does too.

5. Is there a positive sequence b_n similar to a_n where $L = \lim_{n\to\infty} \frac{a_n}{b_n}$ is a finite positive number ($0 < L < \infty$)? If there is, then both series behave the same way. If $L = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $L = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

6. Does the series contain factorials or constants raised to the n-th power? If it does, compute $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$. If $L < 1$, the series converges absolutely. If $L > 1$, the series diverges. If $L = 1$, the test is inconclusive.

7. Is the entire expression raised to the n-th power? If it is, compute $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$. If $L < 1$, the series converges absolutely. If $L > 1$, the series diverges. If $L = 1$, the test is inconclusive.

8. Is the series alternating in the form $\sum(-1)^n b_n$ or $\sum(-1)^{n-1} b_n$ where $b_n > 0$? If it is, check if $b_{n+1} \leq b_n$ and $\lim_{n\to\infty} b_n = 0$. If both conditions hold, the series converges.

Note: After **3**, it is more inspection. With practice (see below), this will become clearer and more natural :).

Practice

Below are 30 series ranging in difficulty. If the series converges, try to find the value it converges to. Not every convergent series has a nice closed form, so if you cannot find the value, just decide whether it converges.

1. $\sum_{n=1}^{\infty} \frac{n^2+1}{3n^2+n}$
2. $\sum_{n=1}^{\infty} \frac{1}{n^4}$
3. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$
4. $\sum_{n=0}^{\infty} 5\left(\frac{2}{3}\right)^n$
5. $\sum_{n=1}^{\infty} \frac{n^2}{n!}$
6. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$
7. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$
8. $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$
9. $\sum_{n=1}^{\infty} \frac{4^n}{3^n}$
10. $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n}}$
11. $\sum_{n=1}^{\infty} \frac{1}{n^2+5n+6}$
12. $\sum_{n=2}^{\infty} \frac{1}{n\ln(n)}$
13. $\sum_{n=1}^{\infty} \frac{n-1}{2n+5}$
14. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+4}$
15. $\sum_{n=1}^{\infty} e^{-n}$
16. $\sum_{n=1}^{\infty} \frac{3^n}{n^3}$
17. $\sum_{n=1}^{\infty} \frac{7}{n}$
18. $\sum_{n=1}^{\infty} \frac{n+2}{n^3-n+5}$
19. $\sum_{n=1}^{\infty} \frac{2^n+3^n}{5^n}$
20. $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{\sqrt{n}}$
21. $\sum_{n=1}^{\infty} \frac{n^3}{e^n}$

22. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+3}}$

23. $\sum_{n=1}^{\infty} \frac{n+5}{n^2+2n}$

24. $\sum_{n=1}^{\infty} \frac{2^n}{n!}$

25. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$

26. $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 3}{n^2+1}$

27. $\sum_{n=1}^{\infty} \frac{1}{2n+3}$

28. $\sum_{n=1}^{\infty} \frac{n!}{5^n}$

29. $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2}\right)$

30. $\sum_{n=1}^{\infty} \frac{2n}{5n+1}$

Answer Key with Justifications

1. **Diverges** via the Divergence Test (Step 2).

$$\lim_{n\to\infty} \frac{n^2+1}{3n^2+n} = \frac{1}{3} \neq 0$$

2. **Converges** immediately as a p-series (Step 1), where $p = 4 > 1$.

3. **Converges** via the Limit Comparison Test (Step 5). Compare it with the convergent p-series $b_n = \frac{\sqrt{n}}{n^2} = \frac{1}{n^{3/2}}$:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\sqrt{n}/(n^2+1)}{1/n^{3/2}} = 1$$

   Since $0 < L < \infty$, both behave the same way.

4. **Converges** immediately as a Geometric Series (Step 1), where $|r| = \frac{2}{3} < 1$. The sum is $\frac{5}{1-2/3} = 15$.

5. **Converges** via the Ratio Test (Step 6) due to the factorial term:

$$L = \lim_{n\to\infty} \left| \frac{(n+1)^2}{(n+1)!} \cdot \frac{n!}{n^2} \right| = \lim_{n\to\infty} \frac{(n+1)^2}{n^2(n+1)} = 0 < 1$$

   The sum is $2e$.

6. **Converges** immediately as a Telescoping Series (Step 1). Rewriting via partial fractions yields $\frac{1}{n} - \frac{1}{n+1}$, and the partial sums collapse down to a total sum of 1.

7. **Converges** via the Alternating Series Test (Step 8). The non-alternating part $b_n = \frac{1}{n}$ is strictly decreasing ($b_{n+1} \leq b_n$) and $\lim_{n\to\infty} b_n = 0$. *(This is the alternating harmonic series, and its sum is* $\ln 2$*).*

8. **Converges** via the Root Test (Step 7) because the whole expression is raised to the n-th power:

$$L = \lim_{n\to\infty} \sqrt[n]{\left| \left(\frac{2n+1}{3n-1} \right)^n \right|} = \lim_{n\to\infty} \frac{2n+1}{3n-1} = \frac{2}{3} < 1$$

9. **Diverges** immediately as a Geometric Series (Step 1), where $|r| = \frac{4}{3} \geq 1$.

10. **Converges** immediately as a p-series (Step 1) by rewriting the radical denominator: $\frac{1}{n \cdot n^{1/2}} = \frac{1}{n^{3/2}}$, where $p = \frac{3}{2} > 1$.

11. **Converges** via the Limit Comparison Test (Step 5). The dominant terms suggest comparing it to the convergent p-series $b_n = \frac{1}{n^2}$, yielding a limit of $L = 1$. *(Can also be solved as a telescoping series via partial fractions expansion, which gives the sum* $\frac{1}{3}$*).*

12. **Diverges** via the Integral Test (Step 3). The function $f(x) = \frac{1}{x\ln(x)}$ is continuous, positive, and decreasing on $[2, \infty)$. Applying u-substitution ($u = \ln(x)$):
$$\int_2^\infty \frac{1}{x\ln(x)}\,dx = \lim_{t\to\infty}\Big[\ln(\ln(x))\Big]_2^t = \infty$$

13. **Diverges** via the Divergence Test (Step 2).
$$\lim_{n\to\infty}\frac{n-1}{2n+5} = \frac{1}{2} \neq 0$$

14. **Converges** via the Alternating Series Test (Step 8). The term $b_n = \frac{n}{n^2+4}$ clearly trends to 0, and a quick derivative check shows $f'(x) = \frac{4-x^2}{(x^2+4)^2} < 0$ for all $x > 2$, meaning it is monotonically decreasing.

15. **Converges** immediately as a Geometric Series (Step 1). Rewrite e^{-n} as $\left(\frac{1}{e}\right)^n$, where $|r| = \frac{1}{e} < 1$ (since $e \approx 2.718$). The sum is $\frac{1/e}{1-1/e} = \frac{1}{e-1}$.

16. **Diverges** via the Divergence Test (Step 2), since $\lim_{n\to\infty}\frac{3^n}{n^3} = \infty \neq 0$. *(The Ratio Test (Step 6) also shows this:)*
$$L = \lim_{n\to\infty}\left|\frac{3^{n+1}}{(n+1)^3}\cdot\frac{n^3}{3^n}\right| = \lim_{n\to\infty}\frac{3n^3}{(n+1)^3} = 3 > 1$$

17. **Diverges** immediately as a constant multiple of the Harmonic Series / p-series (Step 1), where $p = 1 \leq 1$.

18. **Converges** via the Limit Comparison Test (Step 5). Compare it with the convergent p-series $b_n = \frac{n}{n^3} = \frac{1}{n^2}$:
$$L = \lim_{n\to\infty}\left(\frac{n+2}{n^3-n+5}\cdot\frac{n^2}{1}\right) = \lim_{n\to\infty}\frac{n^3+2n^2}{n^3-n+5} = 1$$
Since $0 < L < \infty$, both behave the same way.

19. **Converges** by splitting the expression into two independent, convergent Geometric Series (Step 1): $\sum(\frac{2}{5})^n$ where $r = \frac{2}{5} < 1$, and $\sum(\frac{3}{5})^n$ where $r = \frac{3}{5} < 1$. The sum of two convergent series is always convergent. The sums are $\frac{2}{3}$ and $\frac{3}{2}$, so the total is $\frac{13}{6}$.

20. **Converges** via the Alternating Series Test (Step 8). Because $\cos(n\pi)$ alternates values sequentially $(-1, 1, -1, 1\ldots)$, rewrite the entire structural term as $\frac{(-1)^n}{\sqrt{n}}$. The positive sequence $b_n = \frac{1}{\sqrt{n}}$ decreases steadily to 0.

21. **Converges** via the Ratio Test (Step 6) due to the exponential denominator:
$$L = \lim_{n\to\infty}\left|\frac{(n+1)^3}{e^{n+1}}\cdot\frac{e^n}{n^3}\right| = \lim_{n\to\infty}\frac{(n+1)^3}{e\cdot n^3} = \frac{1}{e} < 1$$

22. **Converges** via the Alternating Series Test (Step 8). The sequence $b_n = \frac{1}{\sqrt{n+3}}$ decreases monotonically and $\lim_{n\to\infty} b_n = 0$.

23. **Diverges** via the Limit Comparison Test (Step 5). Dominating terms match the divergent Harmonic series $b_n = \frac{n}{n^2} = \frac{1}{n}$. Calculating the limit gives $L = 1$, so both diverge.

24. **Converges** via the Ratio Test (Step 6) due to the rapidly growing factorial denominator:

$$L = \lim_{n\to\infty} \left| \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \right| = \lim_{n\to\infty} \frac{2}{n+1} = 0 < 1$$

The sum is $e^2 - 1$.

25. **Converges** via the Integral Test (Step 3). The function $f(x) = \frac{\ln(x)}{x^2}$ is continuous, positive, and decreasing for $x > e$. Using Integration by Parts ($\int u\, dv$):

$$\int_1^\infty \frac{\ln(x)}{x^2}\, dx = \lim_{t\to\infty} \left[-\frac{\ln(x)}{x} - \frac{1}{x} \right]_1^t = 0 - (-1) = 1$$

(Note: the integral equals 1*, but this is not the value of the series. The Integral Test only tells you that the series converges.)*

26. **Converges** via the Alternating Series Test (Step 8), where $b_n = \frac{3}{n^2+1}$ clearly drops to 0. *(Alternatively, it converges absolutely by Direct Comparison (Step 4) because* $\left| \frac{(-1)^n \cdot 3}{n^2+1} \right| < \frac{3}{n^2}$*, which is a convergent p-series).*

27. **Diverges** via the Limit Comparison Test (Step 5) comparing it directly to the divergent harmonic series $b_n = \frac{1}{n}$, which evaluates to a finite limit of $L = \frac{1}{2}$.

28. **Diverges** via the Divergence Test (Step 2), since factorials always outgrow standard exponentials: $\lim_{n\to\infty} \frac{n!}{5^n} = \infty \neq 0$. *(The Ratio Test (Step 6) agrees:)*

$$L = \lim_{n\to\infty} \left| \frac{(n+1)!}{5^{n+1}} \cdot \frac{5^n}{n!} \right| = \lim_{n\to\infty} \frac{n+1}{5} = \infty > 1$$

29. **Converges** immediately as a Telescoping Series (Step 1). Writing out terms reveals that the running partial sum collapses down cleanly:

$$S_n = \left(1 + \frac{1}{2}\right) - \frac{1}{n+1} - \frac{1}{n+2}$$

Taking the limit as $n \to \infty$ leaves a final convergent sum of $\frac{3}{2}$.

30. **Diverges** via the Divergence Test (Step 2).

$$\lim_{n\to\infty} \frac{2n}{5n+1} = \frac{2}{5} \neq 0$$